

Reducing Overfitting in Traffic Sign Classification Using Dropout in Neural Networks

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1. Introduction

Convolutional Neural Networks (CNNs) are widely used for image classification tasks due to their ability to automatically learn hierarchical feature representations. In real-world applications such as autonomous driving, accurate image classification is critical. One example is traffic sign recognition, where models must correctly identify signs under varying lighting, angles, and occlusions.

However, a major challenge in training neural networks is overfitting. Overfitting occurs when a model learns patterns specific to the training data but fails to generalise to unseen data. This leads to high training accuracy but poor performance on test data.

This tutorial focuses on a widely used regularisation technique known as dropout and demonstrates how it improves generalisation in a CNN trained on the German Traffic Sign Recognition Benchmark. The goal is to compare model performance with and without dropout and analyse its effect on both loss behaviour and kernel weight distributions.

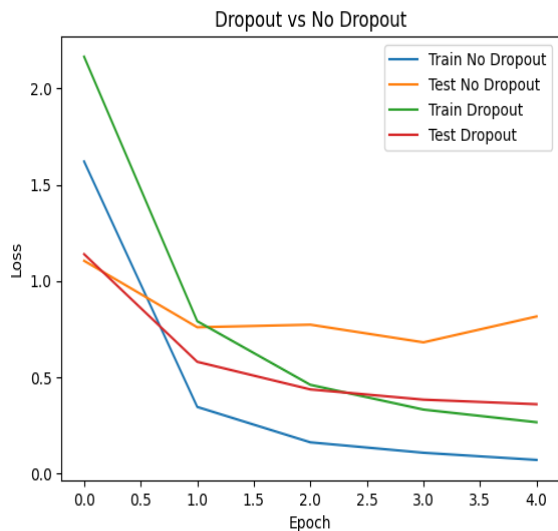
2. Background: Dropout in Neural Networks

Dropout is a regularisation technique introduced to reduce overfitting in deep neural networks. During training, dropout randomly “drops” (sets to zero) a fraction of neurons in a layer with a specified probability.

For example, with a dropout rate of 0.5, each neuron has a 50% chance of being ignored during each training step. This prevents the network from relying too heavily on specific neurons and encourages it to learn more robust and distributed representations.

At inference time, dropout is disabled, and all neurons are used. However, their outputs are scaled appropriately to maintain consistency.

The key intuition behind dropout is Instead of learning one large network, dropout effectively trains an ensemble of many smaller networks, improving generalisation.



3. Dataset: Traffic Sign Recognition

The German Traffic Sign Recognition Benchmark is a widely used dataset for traffic sign classification. It contains images of German road signs belonging to 43 different classes.

Key characteristics:

- Over 50,000 images
- 43 categories (e.g. speed limits, warnings, prohibitions)
- Real-world variations (lighting, angle, background)

This dataset is well-suited for CNNs because:

- Images contain strong visual patterns (shapes, colours, symbols)
- Features can be learned hierarchically

The dataset is well-suited for CNN-based approaches because traffic signs contain strong visual patterns such as distinct shapes, colours, and symbols. CNNs can effectively learn these features through convolutional filters and hierarchical representation learning.

4. Methodology

4.1 Model Architecture

Two CNN models were implemented using PyTorch:

1. Baseline Model (No Dropout)
2. Regularised Model (With Dropout)

Both models share the same architecture:

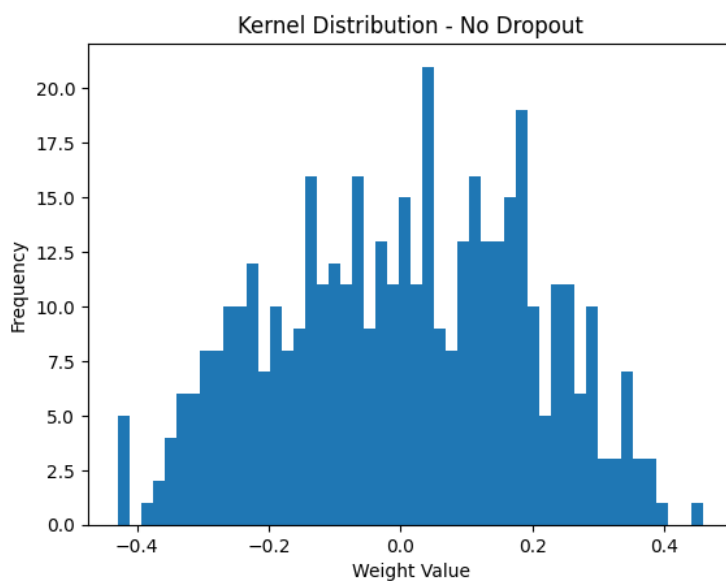
- Two convolutional layers with ReLU activation
- Max pooling layers to reduce spatial dimensions
- Fully connected classifier

The only difference is that the second model includes a dropout layer ($p = 0.5$) before the final classification layer.

4.2 Training Setup

- Loss function: Cross-Entropy Loss
- Optimiser: Adam
- Batch size: 64
- Epochs: 5

Both models were trained on the same dataset under identical conditions to ensure a fair comparison.



5. Results

5.1 Training and Test Loss

The first key result compares training and test loss across epochs for both models.

The model without dropout shows a rapid decrease in training loss, indicating that it is learning the training data effectively. However, the test loss does not decrease at the same rate, suggesting that the model is beginning to overfit.

In contrast, the model with dropout shows a slower reduction in training loss but achieves a more consistent test loss. The gap between training and test performance is smaller, indicating improved generalisation

This demonstrates that dropout helps prevent the model from memorising the training data.

5.2 Kernel Weight Distributions

To further analyse model behaviour, the distributions of kernel weights in the first convolutional layer were examined.

The baseline model shows a wider distribution of weights, with many high-magnitude values. This suggests that the model has learned highly specific features, which may not generalise well to new data. Such extreme weights are often a sign of overfitting.

In contrast, the dropout model exhibits a more concentrated distribution of weights. The values are more balanced, with fewer extreme magnitudes. This indicates that the model is learning more general features rather than memorising specific patterns.

Observations:

- No Dropout Model:
 - Wider distribution of weights
 - More extreme values (high magnitude weights)
 - Indicates over-specialisation
- Dropout Model:
 - More concentrated distribution
 - Fewer extreme values
 - Suggests more balanced learning

These results indicate that dropout constrains the model, preventing weights from becoming overly large and encouraging more stable representations.

5.3 Kernel Statistics

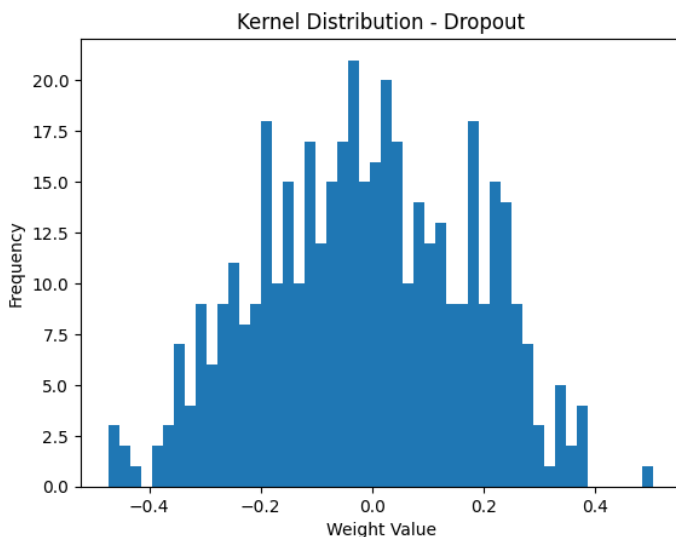
Statistical analysis of kernel weights provides further evidence of the regularising effect of dropout. Key metrics such as mean, standard deviation, and range (maximum and minimum values) were examined.

The dropout model generally shows:

- Lower standard deviation (reduced variance)

- Smaller range of values
- Fewer extreme weights

These characteristics suggest that the model is less prone to overfitting and maintains more stable parameter values. By limiting excessive growth in weights, dropout helps ensure that the model remains well-behaved during training.



6. Discussion

The results clearly demonstrate that dropout is an effective regularisation technique for reducing overfitting in CNNs.

Why Does Dropout Work?

Dropout improves generalisation through several mechanisms:

- It prevents co-adaptation of neurons, ensuring that no single neuron becomes overly important
- It encourages redundancy in feature learning, making the network more robust
- It simulates ensemble learning by training multiple sub-networks
- It introduces noise during training, which acts as a form of regularisation

These properties make dropout particularly useful in complex tasks such as traffic sign recognition, where models must handle diverse and unpredictable inputs.

Trade-offs

While dropout provides significant benefits, it also introduces some trade-offs:

- Slower convergence during training
- Slightly higher training loss
- Increased training time in some cases

However, these drawbacks are generally acceptable given the improvement in test performance and generalisation ability. In safety-critical applications, generalisation is far more important than achieving the lowest possible training loss.

7. Ethical Considerations

Machine learning models used in transport systems, such as traffic sign recognition, can have significant real-world consequences. Errors in classification may lead to unsafe decisions in autonomous vehicles.

Using techniques like dropout to improve generalisation is therefore not only a technical improvement but also an ethical necessity. Robust models reduce the risk of failure in unpredictable real-world scenarios.

Additionally, datasets must be representative to avoid bias. If a model is trained only on specific environments, it may perform poorly in others, potentially leading to unfair or unsafe outcomes.

8. Conclusion

This report explored the role of dropout in reducing overfitting in convolutional neural networks, using a traffic sign classification task as a case study.

The key findings are:

- Dropout reduces the gap between training and test loss
- It leads to more balanced and stable kernel weight distributions
- It improves the model's ability to generalise to unseen data

Although dropout slightly slows down training, its benefits in improving robustness and reliability make it a valuable technique. This is especially important in safety-critical applications such as autonomous driving, where model performance must remain consistent under varying conditions.

In conclusion, dropout is a simple yet powerful method for enhancing the performance of neural networks and should be considered an essential component in modern deep learning pipelines.

9. References

- Dropout: A Simple Way to Prevent Neural Networks from Overfitting
- PyTorch Documentation
- German Traffic Sign Recognition Benchmark Dataset description

GitHub Link: <https://github.com/VENKATESWARLU143-CREATE/Mechine-Learning-Project-.git>